

AUDIO MACHINE LEARNING · DEMO NOTEBOOK

# Foundation Models for Acoustic Scene Analysis

Classifying where a sound was recorded — Restaurant, Street, Office, Car — using  
**BEATs**, **WavLM** and **HuBERT** from Hugging Face.

Pull models & data from Hugging Face → embed → classify

## THE TASK

# What is acoustic scene analysis?

Given a short audio clip, predict the **environment** it was recorded in — not individual sounds, but the whole "scene".



### Restaurant

chatter, cutlery,  
ambience



### Street

traffic, people outside



### Office

keyboards, low talk, AC  
hum



### Car

engine, road noise

Useful for context-aware apps, smart assistants, hearing aids, robotics, and meeting tools that adapt to surroundings.

## THE APPROACH

# Use pre-trained audio models as feature extractors

No training from scratch. Take models already trained on huge audio collections, freeze them, and read out one vector per clip. Then train a tiny classifier on top — a **linear probe**.



**Why it matters:** the heavy model runs only in inference — the part we train is cheap and fast.

## THE MODELS

# Three audio foundation models

| MODEL         | WHAT IT IS                                                                                  | HUGGING FACE SOURCE                                             |
|---------------|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| <b>BEATs</b>  | Audio pre-training with acoustic tokenizers (Microsoft). Strong on general sounds & scenes. | <code>microsoft/unilm</code> + <code>lpepino/beats_ckpts</code> |
| <b>WavLM</b>  | Speech model (Microsoft) that also models background & overlapping noise.                   | <code>microsoft/wavlm-base-plus</code>                          |
| <b>HuBERT</b> | Self-supervised speech model (Meta).                                                        | <code>facebook/hubert-base-ls960</code>                         |

All ~90 M parameters · all output 768-dim embeddings · all pulled straight from the Hub. BEATs isn't in `transformers`, so we fetch Microsoft's reference code + a community checkpoint.

# TUT Acoustic Scenes 2017 — from Hugging Face

## Real recordings

10-second clips of real environments. The dataset's classes map cleanly onto ours:

Restaurant ← `cafe/restaurant`

Street ← `city_center`

Office ← `office`

Car ← `car`

## Smart, lightweight download

Full set is ~12 GB. We don't need it all.

- Download the tiny label CSV first
- Pick ~25 clips per scene
- Fetch **only those files** — ~150 MB total

repo: `Mahia/TUT2017` via `hf_hub_download`

# End-to-end in 9 steps

## 1–2 · Setup

Install deps, configure scenes & sample rate.

## 3–4 · Data

Download clips from HF, load as 16 kHz mono.

## 5 · Models

Load BEATs, WavLM, HuBERT behind one interface.

## 6 · Embed

One model at a time, cache vectors to disk.

## 7–8 · Train + Compare

Linear probe, accuracy, confusion matrices.

## 9 · Predict

`predict_scene(path)` on any new clip.

**Memory-safe by design:** only one ~90 M model is in RAM at a time; embeddings are cached so re-runs are instant.

## PERFORMANCE

# Runs on a laptop CPU — no GPU needed

Because the models are only used for inference on short clips, a normal CPU is enough. Measured on a Ryzen 7 laptop (no CUDA):

**50s**

**BEATs**

embed 100 clips

**69s**

**WavLM**

embed 100 clips

**65s**

**HuBERT**

embed 100 clips

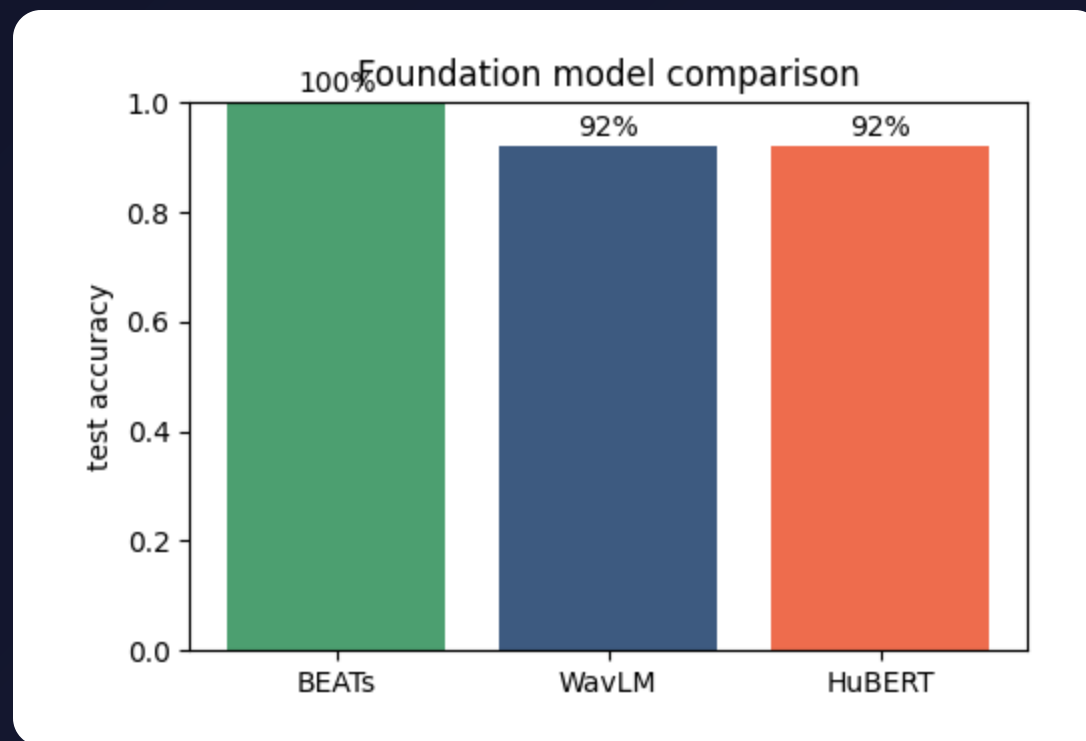
≈ **1 second per clip** per model. Whole notebook end-to-end: **~8 minutes**, mostly first-time downloads. A GPU only helps if you later *fine-tune* the backbones.

## RESULTS

# Accuracy on held-out test clips

| MODEL  | TEST ACCURACY |
|--------|---------------|
| BEATs  | 100%          |
| WavLM  | 92%           |
| HuBERT | 92%           |

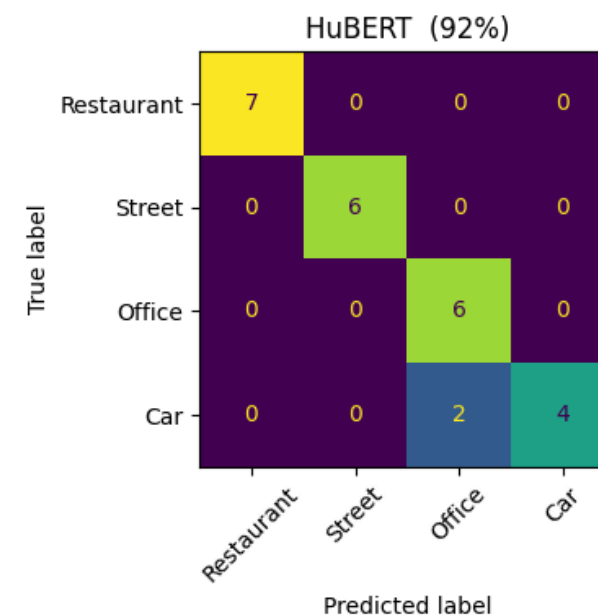
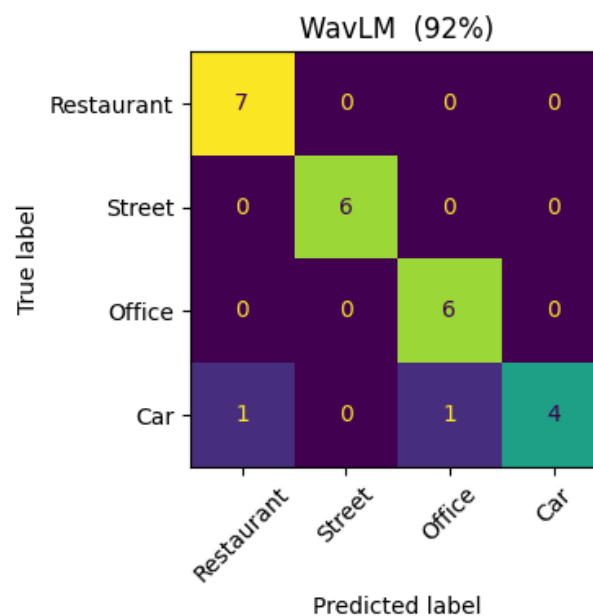
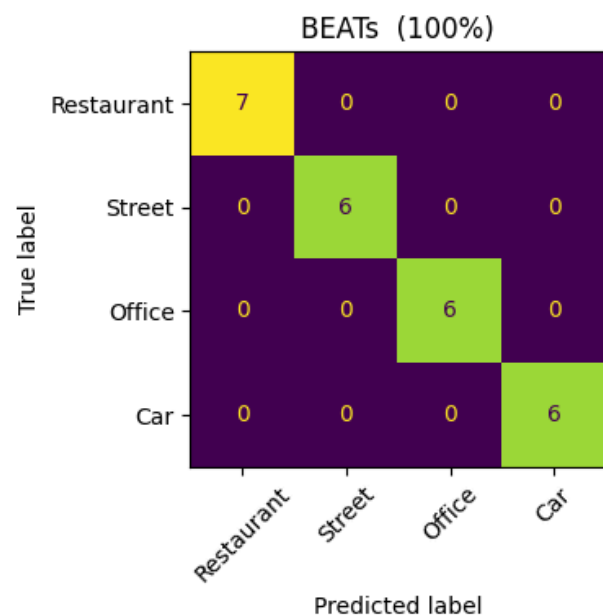
A simple linear classifier on frozen embeddings already separates the four scenes very well — evidence the foundation models capture rich acoustic structure.





## RESULTS

# Where do the models confuse scenes?



BEATs gets every test clip right. WavLM & HuBERT occasionally miss **Car** (engine drone can resemble other steady backgrounds) — the only real source of error.

# Classifying a brand-new clip

```
pred, proba = predict_scene("some_clip.wav")

true scene : Restaurant
predicted  : Restaurant (model: BEATs)
confidence : {'Restaurant': '99.8%', 'Office': '0.1%',
              'Street': '0.1%', 'Car': '0.0%'}
```

One helper turns *any* audio file into a scene prediction with calibrated-looking probabilities — ready to drop into an app.

## TAKEAWAYS

# What this demonstrates

- Foundation models + linear probe is a fast, low-cost recipe for audio classification.
- BEATs — built for general audio — edged out the speech-focused WavLM/HuBERT on scenes.
- Everything (models *and* data) is pulled straight from Hugging Face.
- Reproducible and CPU-friendly: no GPU, no big infra, ~8 minutes start to finish.

## NEXT STEPS

# How to push accuracy further

### More & richer data

Raise clips per class; average several windows per clip instead of one mean-pool.

### Stronger features

Try larger checkpoints (e.g. `wavlm-large`) or concatenate all three models' embeddings.

### Better head

Swap logistic regression for a small MLP; add calibration.

### Fine-tune

Unfreeze the backbone end-to-end — the step that benefits from a GPU.

THANK YOU

# Questions?

Notebook: `notebooks/acoustic_scene_analysis.ipynb`

Models & data: Hugging Face Hub · BEATs · WavLM · HuBERT · TUT Acoustic  
Scenes 2017

Run it top to bottom — results in ~8 minutes on CPU